

SELinux the Easy Way

A Practical Guide for QA Engineers

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Agenda

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What's SELinux?

SELinux (Security-Enhanced Linux) is a security architecture for Linux systems that provides Mandatory Access Control (MAC).

- **MAC vs. DAC:** Unlike traditional Linux permissions (Discretionary Access Control) where users control their files, MAC enforces policies set by the administrator.
- **Default Deny:** If there isn't an explicit rule allowing an action, it is blocked.
- Originally developed by the NSA, now integrated into the mainline Linux kernel.
- Implemented via Linux Security Modules (LSM).

Why SELinux?

- **Containment:** If a service (like a web server) is compromised, SELinux prevents the attacker from accessing the rest of the system or other services.
- **Privilege Escalation Prevention:** Even an exploit running as `root` can be contained if SELinux policies restrict what that specific `root` process can do.
- **Zero-Day Protection:** Helps mitigate the impact of unknown vulnerabilities by enforcing strict behavioral boundaries.
- **Compliance:** Often required by security standards and enterprise environments.

Why QA Cares About SELinux

- SELinux is the **default MAC** on SLES 16, Leap 16, and Tumbleweed
- Customers run SELinux in **Enforcing** mode — so must QA
- SELinux denials look like permission errors — easy to misdiagnose
- A proper bug report with SELinux context saves developers **hours**

Happens every time

"It works on my machine" —
because you had SELinux
disabled.



SELinux Modes



Mode	Behavior	When to use
Enforcing	Blocks + logs denials	Always (production & testing)
Permissive	Logs but does NOT block	Diagnosing SELinux issues only
Disabled	SELinux not initialized	Never (requires reboot + full relabel to re-enable)

```
$ sestatus
```

```
$ getenforce          # Check current mode  
Enforcing
```

```
$ sudo setenforce 0    # Switch to Permissive (temporary)
```

```
$ sudo setenforce 1    # Back to Enforcing
```

```
# Permanent: edit /etc/selinux/config and reboot  
SELINUX=enforcing
```

⚙️ **How SELinux Works** (see link for picture)

SELinux acts as a gatekeeper between a **Subject** (usually a process) and an **Object** (a file, directory, port, etc.).

1. **Subject Request:** A process requests an action (e.g., `read`) on an object (e.g., a file).
2. **DAC Check:** Standard Linux permissions (Owner/Group/Others) are checked first. If this fails, access is denied immediately.

but there's more ...

⚙️ How SELinux Works

3. **MAC Check:** If DAC passes, SELinux steps in. It looks at the **scontext** (source context — the process) and the **tcontext** (target context — the file).
4. **Policy Evaluation:** The SELinux policy is consulted to see if there is an explicit rule allowing the **scontext** to perform the requested action on the **tcontext**.
5. **Decision:** Allowed if a rule exists, blocked otherwise (Default Deny).

see also [This quick intro](#)

Policies: The Rulebook

A **policy** is the set of rules that defines what each domain (process type) is allowed to do.

Policy types (set in `/etc/selinux/config`):

Policy	Description	Use case
targeted	Only known services are confined, everything else runs unconfined	Default on SLES 16 / Tumbleweed / Leap 16
minimum	Like targeted but fewer services confined	Minimal environments (not shipped on SUSE)
mls	Full Multi-Level Security	Government/military only

Policies are modular — built from independent `.pp` modules:

```
# List loaded policy modules
$ sudo semodule -l | head
abrt
account-utils
accountsd
acct
afs

# Install a custom module
$ sudo semodule -i my_fix.pp

# Remove a module
$ sudo semodule -r my_fix
```

SLES 16 ships **440+ policy modules** out of the box (based on Fedora upstream policy with SUSE-specific patches). Custom modules are how you extend the policy.

`audit2allow -M` generates them.

```
# See specific module source (if selinux-policy-devel installed)
sudo zypper install selinux-policy-devel
ls /usr/share/selinux/devel/

# Extract a loaded module to CIL (text) format
sudo semodule -E ftpd          # writes ftpd.cil in cwd
sedismod ftpd.pp              # disassemble .pp binary policy

# Or use sesearch to query rules directly
sudo zypper install setools-console
sesearch --allow -s ftpd_t      # all allow rules for ftpd
sesearch --allow -t etc_t      # all rules targeting etc_t
```

Policy Rules: A Concrete Example

A Type Enforcement (TE) rule looks like this:

```
allow  httpd_t  httpd_sys_content_t : file { read getattr open };  
-----  
      source    target      class      permissions  
      domain    type  
└─ "allow this to happen"
```

Read it as: "A process running in the `httpd_t` domain is allowed to `read`, `getattr`, and `open` files labeled `httpd_sys_content_t`."

Policy Rules: A Concrete Example

You can query existing rules with `sesearch` :

```
# What can httpd_t read?
$ sesearch --allow -s httpd_t -t httpd_sys_content_t -c file
allow httpd_t httpd_sys_content_t:file { getattr ioctl lock map open read };

# What can write to files labeled httpd_log_t?
$ sesearch --allow -t httpd_log_t -c file -p write
allow httpd_t httpd_log_t:file { append create write ... };
```

If no rule exists → **denied**. That's the "default deny" principle.

Where Do Policies Live? (on SUSE)

```
/etc/selinux/
├── config                # SELINUX=enforcing, SELINUXTYPE=targeted
└── targeted/
    ├── policy/policy.33  # compiled binary policy (loaded by kernel)
    ├── contexts/         # default contexts for logins, services, etc.
    └── modules/          # installed policy modules (recently migrated here)

/usr/share/selinux/targeted/  # source .pp modules shipped by packages
```

Key packages (zypper):

Package	Contents
<code>selinux-policy</code>	Base policy, file contexts, core types
<code>selinux-policy-targeted</code>	The targeted policy variant (440+ modules)
<code>container-selinux</code>	Policy for Podman/Docker containers
<code>policycoreutils-python-utils</code>	<code>semanage</code> , <code>audit2allow</code> , <code>audit2why</code>

SUSE-specific: policy modules recently migrated from `/var/lib/selinux` to `/etc/selinux` so they're covered by BTRFS snapshots and rollbacks.

What is a Label?

Every object in the system has a **security context** (label).

A label is made of 4 parts: user, role, type, level

```
system_u : system_r : httpd_t : s0:c1,c2  
  user      role      type      level
```

Label Fields Explained : User

User (`system_u` , `unconfined_u` , `staff_u`)

- Used by RBAC (Role-Based Access Control) policies.
- [Maps Linux users to SELinux users](#). Controls which roles are available to a user.
- `system_u` = system daemons, `unconfined_u` = regular users (no restrictions)

Label Fields Explained : Role

Role (`system_r` , `object_r` , `unconfined_r`)

- Also used by **RBAC**.
- Bridges users to types — dictates which domains (types) a user can transition into.
- `system_r` = system processes, `object_r` = files/passive objects

Label Fields Explained: Type

Type (`httpd_t` , `user_home_t` , `tmp_t`)

- Used by TE (Type Enforcement) policies.
- **The field you'll work with 95% of the time.** In "targeted" policy, almost all rules check this.
- Processes have a **domain** (`httpd_t`), files have a **type** (`httpd_sys_content_t`).
- Rule format: `allow <domain> <type> : <class> { <permissions> };`

Label Fields Explained: Level

Level (`s0` , `s0:c1,c2`)

- Used by **MLS/MCS (Multi-Level / Multi-Category Security)** policies.
- Defines sensitivity and categories. Crucial for **Container isolation** (e.g., Podman gives each container a unique category like `c1,c2`).

👁️ Viewing Labels

```
# Files
$ ls -Z /var/www/html/index.html
system_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html

# Processes
$ ps -eZ | grep nginx
system_u:system_r:httpd_t:s0      1234 ?    00:00:00 nginx

# Your own context
$ id -Z
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

Rule: if a process in domain `X_t` tries to access a file of type `Y_t`, SELinux checks the policy for an `allow X_t Y_t` rule.

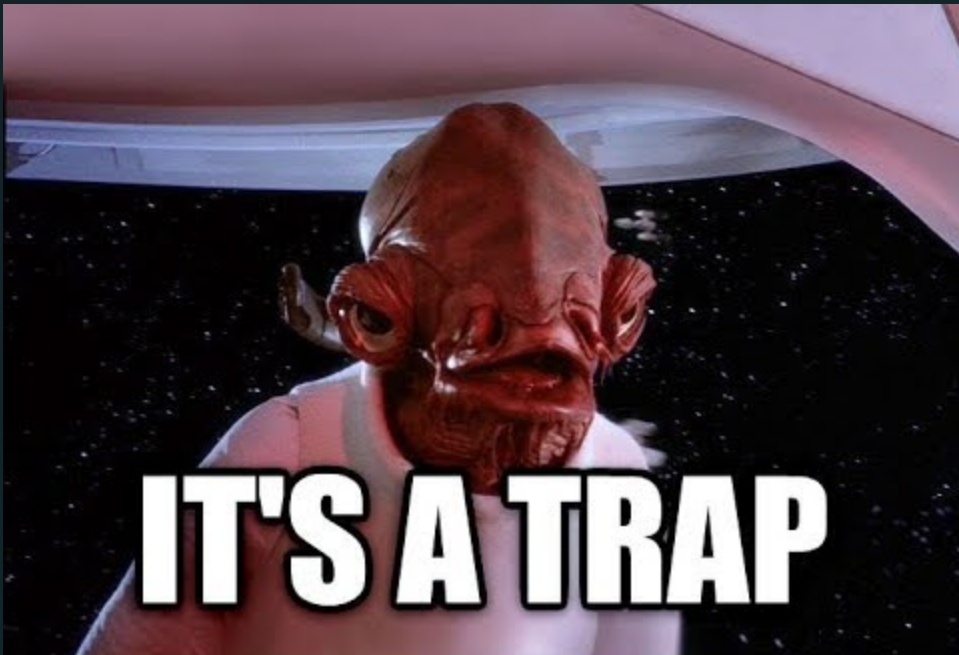
Changing Labels

Tool	Permanent?	Use case
<code>chcon -t httpd_sys_content_t file</code>	No (lost on relabel)	Quick testing
<code>semanage fcontext -a -t httpd_sys_content_t '/srv/web(/.*)?'</code> + <code>restorecon -Rv /srv/web</code>	Yes	Production fix

QA tip: always use `semanage fcontext` + `restorecon` in test procedures. `chcon` changes silently vanish after `restorecon -R` or a full relabel.

⚠ The mv/cp Trap

```
# when a script creates a test config in temp directory  
$ echo "server {}" > /tmp/mysite.conf  
$ ls -Z /tmp/mysite.conf  
unconfined_u:object_r:user_tmp_t:s0 /tmp/mysite.conf
```



⚠ The mv/cp Trap

COPY creates a new file inode → file will be labeled by context rules

```
$ cp /tmp/mysite.conf /etc/nginx/conf.d/  
$ ls -Z /etc/nginx/conf.d/mysite.conf  
unconfined_u:object_r:httpd_config_t:s0      # correct!
```

MOVE keeps original label

```
$ mv /tmp/mysite.conf /etc/nginx/conf.d/  
$ ls -Z /etc/nginx/conf.d/mysite.conf  
unconfined_u:object_r:user_tmp_t:s0          # WRONG! nginx can't read this
```

Fix: `restorecon -v /etc/nginx/conf.d/mysite.conf`

QA Angle: Labels in Bug Reports

When filing a bug where SELinux might be involved, **always include:**

```
# 1. Label of the file/directory that failed
ls -Z /path/to/file

# 2. Label of the process that got denied
ps -eZ | grep <process_name>

# 3. What label the file SHOULD have
restorecon -n -v /path/to/file

# 4. Current SELinux mode
getenforce
```

A bug report that says "permission denied" without this info is **incomplete**.

● Booleans: Policy Switches

Modify policy *behavior* without writing custom rules.

```
# List all booleans related to httpd
$ getsebool -a | grep httpd
httpd_can_network_connect --> off
httpd_can_sendmail --> off
httpd_enable_homedirs --> off

# Allow Apache to make network connections
$ sudo setsebool -P httpd_can_network_connect on
```

The **-P** flag = persistent across reboots. Without it, the change is lost on restart.

Port Labeling

SELinux controls which processes can **bind** to which ports.

```
# See what ports SSH is allowed to use
$ sudo semanage port -l | grep ssh
ssh_port_t                tcp          22

# Move SSH to port 2222 → it will FAIL to start
# Fix: add the port to the policy
$ sudo semanage port -a -t ssh_port_t -p tcp 2222

# Verify
$ sudo semanage port -l | grep ssh
ssh_port_t                tcp          2222, 22
```

Same applies to any service using a non-standard port.

File Context Rules

The **permanent** way to manage labels for custom paths:

```
# Tell SELinux: everything under /srv/myapp is web content
$ sudo semanage fcontext -a -t httpd_sys_content_t '/srv/myapp(/.*)?'

# Apply the rule to existing files
$ sudo restorecon -Rv /srv/myapp
Relabeled /srv/myapp from unconfined_u:object_r:default_t:s0
  to unconfined_u:object_r:httpd_sys_content_t:s0

# Verify
$ ls -Z /srv/myapp/
system_u:object_r:httpd_sys_content_t:s0 index.html
```

✓ QA Angle: Testing with SELinux

Rules for QA test environments:

1. **Always test in Enforcing mode** — matches customer environments
2. Use Permissive **only** to diagnose, then switch back immediately
3. Never put `setenforce 0` in test setup scripts
4. After installing/upgrading a package, run `restorecon -R` on its paths
5. If a test fails with "Permission denied", check SELinux **before** filing a DAC bug

Where to Look for Denials

```
# The audit log (primary source)
$ sudo ausearch -m AVC -ts recent

# Or directly
$ sudo grep "avc: denied" /var/log/audit/audit.log

# Via journald
$ sudo journalctl -t audit --since "10 minutes ago" | grep AVC
```

No denials showing? Check:

- Is `auditd` running? (`systemctl status auditd`)
- Is SELinux actually in Enforcing? (`getenforce`)
- Some denials are "dontaudit" — use `semodule -DB` to disable dontaudit rules and rebuild policy (re-enable with `semodule -B`)

Anatomy of a Denial

```
type=AVC msg=audit(1716000000.123:456): avc:  denied  { read }
    for pid=1234 comm="nginx" name="mysite.conf"
    dev="sda1" ino=5678
    scontext=system_u:system_r:httpd_t:s0
    tcontext=unconfined_u:object_r:user_tmp_t:s0
    tclass=file permissive=0
```

Field	Meaning
<code>{ read }</code>	Which action was denied
<code>comm="nginx"</code>	Which program tried
<code>scontext=...httpd_t...</code>	Process domain (subject)
<code>tcontext=...user_tmp_t...</code>	File type (target)
<code>tclass=file</code>	Object class

⚡ Quick Check: is it a SELinux issue?

```
# Step 1: Something fails. Is SELinux blocking it?  
$ sudo setenforce 0          # Permissive mode  
  
# Step 2: Retry the operation  
$ sudo systemctl restart nginx  
# If it works now → SELinux was blocking it  
  
# Step 3: IMMEDIATELY switch back  
$ sudo setenforce 1  
  
# Step 4: Find the denial  
$ sudo ausearch -m AVC -ts recent
```

Never leave a system in Permissive mode after testing. Automate: set a timer or use `at` to switch back.

audit2why & audit2allow

```
# WHY was it denied?
$ sudo ausearch -m AVC -ts recent | audit2why
Was caused by: Missing type enforcement (TE) allow rule.
Allow rules: allow httpd_t user_tmp_t:file read;

# WHAT rule would fix it?
$ sudo ausearch -m AVC -ts recent | audit2allow
#===== httpd_t =====
allow httpd_t user_tmp_t:file read;

# Generate a loadable policy module
$ sudo ausearch -m AVC -ts recent | audit2allow -M my_nginx_fix
$ sudo semodule -i my_nginx_fix.pp
```

Warning: `audit2allow` may suggest overly broad rules. Always check what it proposes before loading.

setroubleshoot / sealert

Human-readable analysis of denials:

```
$ sudo sealert -a /var/log/audit/audit.log

SELinux is preventing nginx from read access on the file mysite.conf.

***** Plugin restorecon (99.5 confidence) suggests *****
If you want to fix the label:
    restorecon -v /etc/nginx/conf.d/mysite.conf

***** Plugin catchall (1.49 confidence) suggests *****
If you want to allow httpd_t to read user_tmp_t files:
    audit2allow -M mypol
```

sealert ranks suggestions by confidence — start from the top.

Container Volume Isolation (MCS)

SELinux uses **MCS** (Multi-Category Security) to isolate containers.

- Each container gets unique category pair: `s0:c123,c456`
- Prevents containers from accessing each other's volumes
- Even if running as same UID

Example Pod security context:

```
spec:
  securityContext:
    seLinuxOptions:
      level: "s0:c0.c1023" # allow broad category range
```

Kubernetes 1.27+: Efficient mount-time labeling — no recursive `chcon` on large volumes. See [K8s efficient relabeling blog](#).

Storage Backend Issues & Fixes

Backend	Issue	Fix
Longhorn	iSCSI <code>dac_override</code> denials (container-selinux v2.189.0+)	Load custom module: <code>echo '(allow iscsid_t self (capability (dac_override)))' > fix.cil && semodule -vi fix.cil</code> KB
local-path-provisioner	SELinux blocks hostPath access	<code>chcon -t container_file_t -R /opt/local-path-provisioner</code> Issue
NFS volumes	Container can't mount NFS	<code>setsebool -P virt_use_nfs on</code>

RKE2 + SELinux References

Official docs:

- [RKE2 SELinux Documentation](#)
- [SUSE RKE2 SELinux Guide](#)
- [Kubernetes Security Contexts](#)

Known issues:

- [RKE2 Known Issues](#) — SELinux module upgrade problems
- [Downstream cluster SELinux setup](#)

QA testing notes:

- Always test with `selinux: true` in RKE2 config
- Check volume mount denials: `ausearch -m AVC | grep mount`

Demo Setup

Prerequisites: openSUSE Tumbleweed with `podman` installed.

```
# From this repo's directory:
$ ./selinux-lab.sh

# This will:
# 1. Create a Tumbleweed distrobox named "selinux-lab"
# 2. Install SELinux tools, nginx, audit utilities
# 3. Copy a sample audit log for offline demos

# Enter the lab:
$ podman start -ai selinux-lab
```

All demo commands in the following slides can be run inside this container. Note: actual enforcement requires a system with SELinux enabled in the kernel (Tumbleweed, SLES 16).

★ Demo: nginx Can't Read Its Config

```
# 1. Create a config file in /tmp
echo 'server { listen 8080; root /usr/share/nginx/html; }' \
    > /tmp/demo.conf

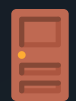
# 2. Move it (not copy!) to nginx config dir
mv /tmp/demo.conf /etc/nginx/conf.d/

# 3. Check the label – it's WRONG
ls -Z /etc/nginx/conf.d/demo.conf
# → user_tmp_t (should be httpd_config_t)

# 4. On a real enforcing system: nginx would fail to start
#     Check the audit log for the AVC denial

# 5. Fix it
sudo restorecon -v /etc/nginx/conf.d/demo.conf
# → Relabeled to httpd_config_t

# 6. Verify
ls -Z /etc/nginx/conf.d/demo.conf
```



Demo: Service on a Non-standard Port

```
# 1. Check which ports httpd_t can bind to
sudo semanage port -l | grep http_port
# → http_port_t      tcp      80, 81, 443, 488, ...

# 2. Try to use port 8888 → SELinux would block it
#     The AVC denial would show:
#     denied { name_bind } for ... scontext=...httpd_t...

# 3. Add port 8888 to the allowed list
sudo semanage port -a -t http_port_t -p tcp 8888

# 4. Verify
sudo semanage port -l | grep http_port
# → http_port_t      tcp      8888, 80, 81, 443, ...

# 5. Now the service can bind to port 8888
```

QA Cheat Sheet

Task	Command
Check SELinux mode	<code>getenforce</code>
View file labels	<code>ls -Z /path/to/file</code>
View process labels	<code>ps -eZ grep <name></code>
Find recent denials	<code>ausearch -m AVC -ts recent</code>
Explain a denial	<code>ausearch -m AVC -ts recent audit2why</code>
Fix a wrong label	<code>restorecon -v /path/to/file</code>
Check expected label	<code>restorecon -n -v /path/to/file</code>
Add permanent label rule	<code>semanage fcontext -a -t <type> '/path(/.*)"'</code>
Toggle boolean	<code>setsebool -P <bool> on</code>

🙏 Thank You!

Questions?

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Lab setup & slides:

https://github.com/ilmanzo/suse_presentations

